

EDUCATION

Bachelor of Arts in Computer Science, Minor in Math

Expected May 2027

- **Relevant Coursework:** Data Structures, Systems Programming, Linear Optimization, Probability, Computer Architecture, Linear Algebra, Differential Equations, Operating Systems, Data Management, Software Meth

EXPERIENCE

Machine Learning Intern

Jul 2025 – Present

Remote

- Designed and implemented a **multi-agent RAG** pipeline to improve multi-hop reasoning accuracy by **10%**
- Boosted multi-hop QA accuracy **5–10 points** by optimizing **Neo4j** vector-store reasoning pipelines
- Scaled model evaluation by implementing automated benchmarks for HotpotQA and 2WikiMultiHopQA

Part-Time Software Engineer

Jun 2025 – Present

East Brunswick, NJ

- Improved website usability by maintaining frontend features with **HTML/CSS** and **JavaScript**
- Raised exam scores by **15–25%** by tutoring **20+** students in calculus, linear algebra, and probability
- Increased platform reliability through site updates, bug fixes, and scheduling improvements

Software Engineering Intern

May 2024 – Aug 2024

Miami, Florida

- Automated **10+** insurance call flows by building a **Python**-based simulation system using REST APIs
- Reduced backend latency by **20%** by redesigning API workflows for multi-practice claims processing
- Accelerated product adoption by incorporating feedback from **5+** dentists into validated workflow improvements

PROJECTS

HackPrinceton Winner – Synthetic Mouse Generator | *Python, Pipelines, MCP*

- Designed and implemented a modular, multi-stage **Python** data pipeline with configurable stages and deterministic outputs to generate synthetic datasets for testing and experimentation
- Built high-throughput data stitching and serialization logic to assemble **2k–5k** records per run and export structured **FASTQ/VCF/CSV** artifacts, improving pipeline reliability and reproducibility

RU-Roomie | *JavaScript, React Native, NativeWind, Node.js, MongoDB, Expo*

- Built a full-stack mobile application using **React Native** and **Node.js** supporting swipe-based matching, real-time filtering, and secure authentication workflows
- Designed and implemented RESTful APIs and optimized **MongoDB** schemas for user profiles, preferences, and messaging, supporting thousands of concurrent users

ProcessSys | *C, Python, Linux*

- Developed a Linux system-call tracing tool in **C** using **ptrace()**, capturing syscall activity across processes for system analysis
- Built a **Python** post-processing pipeline to parse trace outputs, map syscall IDs, and generate frequency reports, reducing debugging time

TraceGuard – API Observability & Failure Simulator | *Python, FastAPI, PostgreSQL, Docker*

- Built an API observability service using **FastAPI** to collect latency, error rates, retries, and response metrics
- Implemented configurable failure injection mechanisms, including timeouts, **5xx** errors, and latency spikes, to validate service resilience

TECHNICAL SKILLS

Languages: Python, Java, C, JavaScript, TypeScript, SQL, MATLAB, HTML/CSS

Frameworks & Libraries: Spring Boot, React.js, Next.js, Node.js, FastAPI, PyTorch, TensorFlow, scikit-learn

Databases, Tools & Platforms: PostgreSQL, MongoDB, Supabase, Git, Linux, Docker, Kubernetes, AWS, Azure

Competencies: QA Testing, Debugging, Agile Development, Cross-Functional Collaboration, OOP